

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA0301

COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO5: Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)

Assessment Tool Weightage: 35%

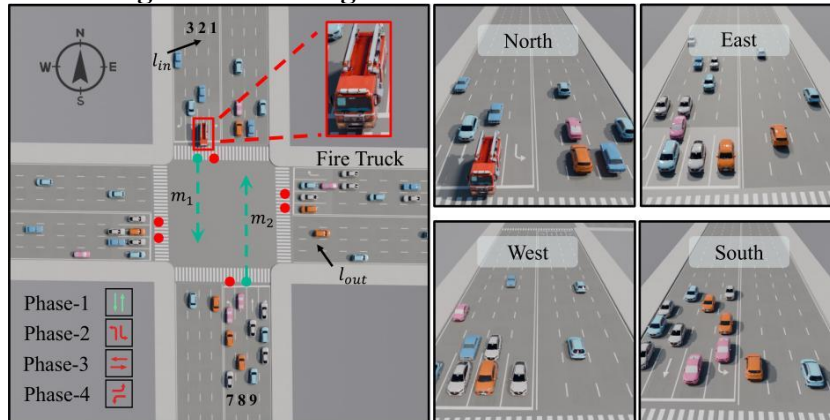
Dr G. A. SENTHIL

QUESTIONS: 1

Total Marks: 50

Annexure A Scenario-Based Assessment

Reinforcement Learning for Traffic Signal Control



Scenario:

A traffic management authority wants to use Reinforcement Learning (RL) to control traffic lights at busy intersections. The main objectives are to reduce traffic congestion, minimize vehicle waiting time, give priority to emergency vehicles such as ambulances and fire trucks, and ensure pedestrian safety.

1. Reinforcement Learning Model

In this system, the **RL agent is the traffic signal controller**. The intersection acts as the environment. The agent continuously observes the traffic conditions, selects a suitable traffic-light action, and receives a reward based on the resulting traffic flow and safety.

Basic workflow:

Traffic Sensors/Cameras → Traffic State → RL Agent → Traffic Signal Action → Traffic Flow → Reward → RL Agent

The system can collect information such as:

- Number of vehicles on each road
- Queue length

- Vehicle waiting time
- Vehicle speed
- Pedestrian presence
- Pedestrian waiting time
- Current traffic-light phase
- Emergency vehicle presence
- Distance of an emergency vehicle from the intersection

Research on RL-based traffic control commonly uses vehicle and pedestrian information together with signal-phase information as part of the state representation. ([MDPI](#))

2. State Space

The **state** represents the current condition of the intersection.

A possible state can be defined as:

State = {vehicle density, queue length, waiting time, pedestrian count, pedestrian waiting time, signal phase, emergency vehicle status}

For example:

- North road: 20 vehicles
- South road: 15 vehicles
- East road: 5 vehicles
- West road: 30 vehicles
- Pedestrians waiting: 10
- Current phase: North-South green
- Emergency vehicle: Present on East road

The RL agent uses this information to decide which traffic-light phase should be activated.

3. Action Space

The **action** represents the decision made by the RL agent.

Possible actions are:

1. Keep the current green signal.
2. Switch to the next signal phase.
3. Give North-South traffic a green signal.
4. Give East-West traffic a green signal.
5. Activate a pedestrian crossing phase.
6. Give priority to an emergency vehicle.

A yellow transition period should be included between conflicting green phases to maintain safety rather than allowing an immediate unsafe change. Traffic-signal RL research also incorporates signal-phase and timing constraints into the action design. ([IET Research](#))

4. Reward Function

The reward function is very important because it tells the RL agent what behavior is desirable.

A suitable reward can be:

Reward = Reduction in vehicle waiting time + reduction in queue length + traffic throughput + pedestrian safety + emergency priority – safety violations

The agent can receive:

- **Positive reward** for reducing vehicle waiting time.
- **Positive reward** for reducing queue length.
- **Positive reward** for allowing more vehicles to pass.
- **Positive reward** for safely allowing pedestrians to cross.
- **Large positive reward** for quickly clearing a path for an emergency vehicle.
- **Negative reward** for increasing congestion.

- **Large negative reward** for pedestrian conflicts or unsafe signal changes.
- **Large negative reward** for delaying an emergency vehicle unnecessarily.

Modern RL traffic-control approaches can use multi-objective rewards that balance vehicle efficiency and pedestrian safety rather than optimizing vehicle throughput alone.

5. Emergency Vehicle Priority

Emergency vehicles require special treatment because their delay can affect life-saving response times.

The system can detect an ambulance, fire truck, or police vehicle using cameras, GPS, connected-vehicle communication, or other sensors.

When an emergency vehicle approaches:

Emergency vehicle detected → Check safe signal transition → Provide priority green →

Emergency vehicle crosses → Return to normal traffic control

For example, if an ambulance is approaching from the East:

- The system detects the ambulance.
- The RL agent checks the current traffic situation.
- Conflicting traffic signals are safely changed.
- The East direction receives priority.
- The ambulance crosses the intersection with minimum delay.
- After the ambulance passes, normal signal operation resumes.

Emergency-vehicle priority can therefore be incorporated into the RL state and reward structure. Research has also investigated RL traffic signal systems that explicitly consider emergency vehicles and pedestrians.

6. Pedestrian Safety

Pedestrian safety must be treated as a **hard safety requirement**, not simply as another traffic optimization objective.

The system can detect:

- Number of pedestrians waiting
- Pedestrian crossing requests
- Pedestrians currently crossing
- Pedestrian crossing time
- Vehicles approaching the pedestrian crossing

The RL agent should not allow conflicting vehicle movements while pedestrians have a protected crossing phase.

For example:

Pedestrian detected → Stop conflicting vehicles → Give pedestrian green → Allow safe crossing → Resume vehicle phase

A large negative reward or safety constraint can be applied if the controller creates a pedestrian-vehicle conflict.

Recent research has specifically investigated deep RL approaches that optimize both pedestrian safety and traffic efficiency

7. Impact on Congestion Reduction

Traditional traffic lights generally use fixed or predefined timings. They may continue giving green time to a road even when that road has very few vehicles.

RL can adapt the signal according to **real-time traffic conditions**.

For example:

Situation	RL Decision
Heavy traffic on North road	Increase North-South green time
Heavy traffic on East road	Give East-West priority
Few vehicles on a road	Reduce unnecessary green time
Many pedestrians waiting	Provide safe pedestrian phase
Ambulance approaching	Give emergency priority
Congestion increasing	Adjust signal timing to reduce queues

This adaptive behavior can reduce:

- Average waiting time
- Queue length
- Unnecessary stops
- Traffic congestion
- Fuel consumption
- Emergency vehicle delay

Studies of pedestrian-aware RL traffic control have reported reductions in waiting time and the number of people waiting compared with some conventional RL baselines, showing the potential of RL for adaptive intersection management. ([MDPI](#))

8. Suitable RL Algorithm

A suitable algorithm for this problem is **Deep Q-Network (DQN)** or an advanced version such as **Dueling DQN**.

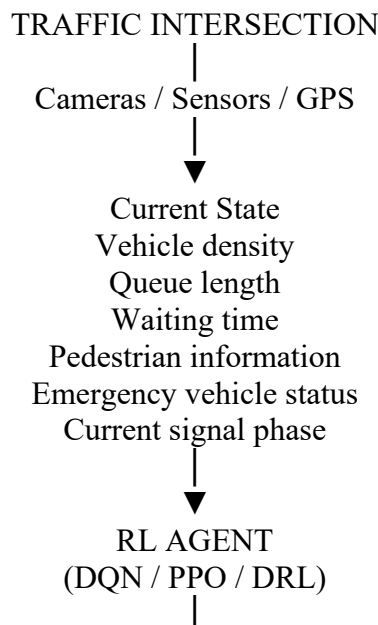
For larger and more complex traffic networks, **PPO** or **Multi-Agent Reinforcement Learning** can also be used.

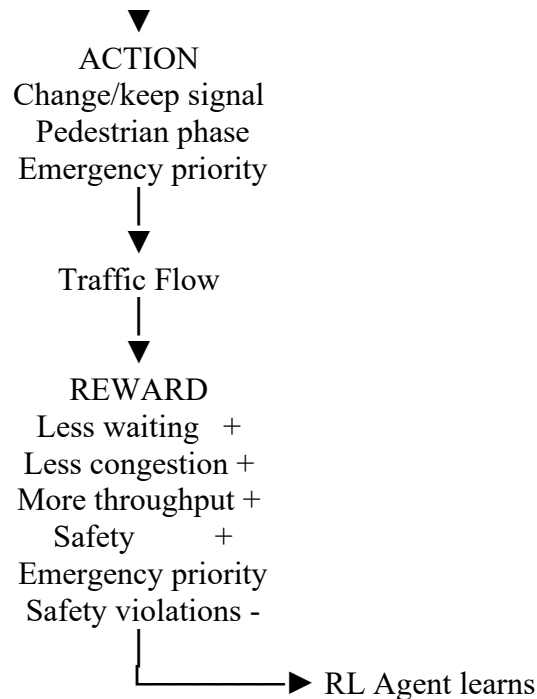
A simple implementation can use:

Environment → **State** → **DQN Agent** → **Action** → **Traffic Simulator** → **Reward** → **Learning**

Traffic simulators such as **SUMO** can be used to train and evaluate the RL controller before deployment in a real intersection. Research commonly uses traffic simulation for testing RL-based signal-control methods.

9. Overall RL Framework





Conclusion

Reinforcement Learning is suitable for intelligent traffic signal control because it can **learn from changing traffic conditions and continuously improve signal decisions**. The state should include traffic density, queues, waiting times, pedestrians, signal phase, and emergency vehicles. The action should control signal phases, while the reward should encourage lower congestion, shorter waiting times, higher throughput, pedestrian safety, and emergency vehicle priority. **Safety constraints and large penalties for unsafe actions are essential** so that congestion reduction never comes at the cost of pedestrian or emergency-vehicle safety.

Hence, an RL-based traffic signal system can provide adaptive, safe, and efficient traffic management compared with purely fixed-time signal control.

Code of Conduct:

*I **Yoga Madha A-192425058** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: